From Erdős to Riemann: A Compute-Ordered Research Agenda for AI-Driven Pure Mathematics

The state of the art in October 2026, an impact-scored map of 26 open problems, and the order in which an AI-only research program should attack them

Prepared for M. Bagus Prabowo

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Abstract

Between May and October 2026, AI systems moved from solving olympiad problems to settling named research conjectures at a rate of several per week. An OpenAI model disproved Erdős's 1946 unit-distance conjecture; Claude helped produce a counterexample to the 87-year-old Jacobian conjecture in dimension three and raised the proven share of Riemann zeta zeros on the critical line from 41.6% to 67.2%; Sendov's and Crouzeix's conjectures were proved with GPT-5.6; and an OpenAI agent swarm of about 10,000 agents produced a Lean-verified answer to the Clay formulation of the Navier–Stokes problem, though only through a forced variant most experts regard as the less important question. Yet on Epoch AI's 68 hardest open Erdős problems the best model has resolved two, and no Millennium Problem has fallen in the form mathematicians actually care about. This paper asks a practical question: if one were to run a frontier pure-mathematics research program using AI alone, which problems should be attacked first, and how should compute be spent from the cheapest wins to the hardest moonshots? We (i) survey the state of the art as of 8 October 2026; (ii) extract five empirical regularities from the record, chief among them that verifiable constructions and counterexamples are cheap while new structural theory is expensive; (iii) define an Impact Score (0–100), a compute tier anchored to observed token costs (10^6 to beyond 10^{11} output tokens), and a Priority Index that combines them with a probability of success; and (iv) score 26 open problems in five tiers, from record-breaking constructions to the Riemann Hypothesis and P vs NP. We close with a phased research program, a verification protocol, and a list of partial-progress targets that let even the hardest problems absorb compute productively.

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1 Introduction

1.1 The question

Mathematics is the cleanest test of machine reasoning there is. A proof is either correct or not, it can be checked by a small program, and the open problems are public, dated, and ranked by a community that has argued about their importance for a century. That is why AI laboratories now treat open problems as their highest-status benchmark, and why 2026 has been the year the dam broke.

This paper takes the perspective of a researcher with access to frontier models and a fixed compute budget, and no human mathematicians on staff. It asks three questions:

1. **Where is the frontier?** What has AI actually resolved, under what checks, and at what cost?
2. **What is cheap and what is expensive?** Which kinds of open problems fall to a few million tokens, and which resist billions?
3. **What should be attacked, in what order?** Given impact and cost, which problems give the most mathematics per token, and how should a program climb from easy wins to moonshots?

1.2 Contributions

- A dated survey of AI in pure mathematics through 8 October 2026, separating claims that are machine-verified, expert-verified, and self-reported (Section 2).
- Five empirical regularities that explain why some problems fall and others do not (Section 2.4).
- A scoring framework: an **Impact Score** $I \in [0, 100]$, a **compute tier** expressed in $\log_{10}$ output tokens and anchored to published runs, and a **Priority Index** Π (Section 3).
- Scored profiles of 26 open problems across five tiers, each with an attack plan, a verifier, and a ladder of partial results (Sections 4–8).
- A phased research program with budgets and a verification and disclosure protocol (Section 10).

1.3 A note on scope and status

The field is moving faster than any survey. One independent tracker counted 1,712 arXiv mathematics papers with substantive AI contributions between 1 March and 20 August 2026, about 14% of submissions by August [21], and OpenAI released 372 claimed results on 6 October 2026, two days before this paper's date [12]. Every problem below is listed as open **to the best of our knowledge on 8 October 2026**; some may already have claimed resolutions in that release or in preprints we have not seen. Scores are the author's structured judgments, not measurements.

2 State of the art (October 2026)

2.1 Milestones

Table 1: Selected milestones. "Lean" means a published formal proof that a proof checker accepts.

Date	Result	System	How it was checked
Jul 2024	IMO silver standard, 28/42	AlphaProof + AlphaGeometry 2	Formal (Lean)
Jul 2025	IMO gold, 35/42	Gemini Deep Think	Graded by IMO coordinators

Date	Result	System	How it was checked
Nov 2025	67 problems explored, several bounds improved	AlphaEvolve (with Tao et al.)	Computational verifiers [7]
Jan 2026	Erdős #728, first Erdős problem resolved autonomously	GPT-5.2 Pro + Aristotle	Lean [3]
Jan 2026	Putnam 2025, 12/12	AxiomProver	Lean [17]
May 2026	9 of 353 open Erdős problems; 44 of 492 OEIS conjectures	AlphaProof Nexus	Lean [4]
May 2026	Erdős unit-distance conjecture (1946) disproved	Internal OpenAI model	Lean; 9-mathematician writeup [9]
Jun 2026	First Proof batch 2: 6–7 of 10 research problems solved by at least one system	GPT-5.5 Pro and academic harnesses	30 blind expert referees [14]
Jul 2026	Jacobian conjecture false in dimension 3 (1939)	Claude Fable 5 with L. Alpöge	Hand-checkable explicit map [10]
Jul–Aug 2026	Crouzeix's conjecture (2004) proved	GPT-5.6 Sol, 16-hour autonomous run	Experts incl. Crouzeix; Lean [18]
Aug 2026	Sendov's conjecture (1958) proved for all degrees	GPT-5.6 Pro	Tao's digestion; Lean [19]
Aug 2026	Zeta zeros on critical line: 41.6% → 67.2%	Unreleased Claude	Experts; Lean [11]
Sep 2026	Navier–Stokes, Clay statement (C) with external force	OpenAI swarm (~10,000 agents)	Lean [5]; disputed relevance [6]
Sep 2026	Every FrontierMath Tier 4 problem solved by some model	GPT-6 Astra / 6.1 Sol	Answer keys [15]
Oct 2026	372 claimed results incl. Kakeya in $\mathbb{R}^4$	Internal OpenAI model	Partly Lean; under review [12]

2.2 Benchmarks: answer keys are saturated, open problems are not

Table 2: The gap between known-answer benchmarks and genuinely open problems.

Benchmark	What it measures	Best result (as of Oct 2026)
FrontierMath Tiers 1–3 (v2)	Hard problems with known answers	about 94% [15]
FrontierMath Tier 4 (v2)	Research-level, known answers	100% (GPT-6.1 Sol) [15]
First Proof, batch 2	10 unpublished research lemmas, refereed blind	6–7 of 10 solved by at least one system; best single system 6/10 [14]
FrontierMath: Open Problems	49 unsolved problems with computer verifiers	5 solved by AI, 5 by humans with AI, 39 open [16]

Benchmark	What it measures	Best result (as of Oct 2026)
FrontierMath: Erdős	68 hard open Erdős problems, Lean proof required	2 of 68 (GPT-6 Astra, pre-release) [20]

The pattern is stark. On problems whose answers exist somewhere, frontier models are at or near ceiling. On problems that resisted professional mathematicians, success rates are between 3% and 20%, and they skew toward the "moderately interesting" end: of Epoch's Open Problems, AI has touched 6 of 22 rated *moderately interesting* but only 1 of 3 rated *breakthrough* [16].

2.3 Four paradigms of AI mathematics

1. **Evolutionary search with a verifier.** A language model writes programs that generate candidate objects; an automatic scorer keeps the best; the loop repeats. AlphaEvolve matched the best known constructions on most of 67 problems and beat them on several [7]; open 8B-parameter models running the same idea (ThetaEvolve) improved bounds too, which means this paradigm is cheap and democratized.
2. **Formal proof search.** A model proposes Lean proofs and the Lean kernel rejects anything wrong. AlphaProof Nexus solved 9 of 353 formalized Erdős problems at a few hundred dollars each [4]; Harmonic's Aristotle and AxiomProver play similar roles. Correctness is guaranteed; *statement fidelity* is not.
3. **Long-horizon natural-language agents.** A single strong model runs for hours with tools, critics and subagents. Crouzeix fell to one 16-hour run [18]; the zeta-zero bound took two Claude Code sessions, about 60 subagents and 31 million output tokens [11].
4. **Massive multi-agent swarms.** Thousands of agents explore in parallel, cross-pollinated by a coordinator. OpenAI's Navier–Stokes run used on the order of 10,000 concurrent agents for 88 hours and about 130 billion output tokens [5].

2.4 Five regularities in the record

R1. Verification asymmetry decides tractability. The problems that fell fastest have answers that are hard to find but easy to check: a counterexample map, a point configuration, a matrix, a Lean term. The Jacobian counterexample was, in Tao's words, "an extremely quick verification" [10].

R2. Constructions beat theory. "LLMs are especially good at constructing things that are very explicit and not as good—for now—in making new theory," as Cao-Labora put it after the Navier–Stokes episode [6]. The September flood of results is dominated by counterexamples and disproofs [21].

R3. Obscurity is not difficulty. In DeepMind's December 2025 sweep of 700 "open" Erdős problems, 212 answers looked plausible, 63 were technically correct, 13 answered the intended question, and only 5 were novel; the authors concluded many problems were open "through obscurity rather than difficulty" [8]. Cheap wins come disproportionately from neglected problems.

R4. Compute buys depth, roughly log-linearly. Published runs span five orders of magnitude, from a 38-minute single prompt [21] to 130 billion tokens [5]. Each step up the ladder has bought a qualitatively harder class of result (Section 3.2).

R5. Fidelity and provenance are the new bottlenecks. Erdős #728 was ambiguous as written [9]; OpenAI's Navier–Stokes proof solved the Clay statement through an external force that mathematicians call a loophole, and a follow-up showed the method cannot remove it [6]. Models also rediscover

literature without citing it, which DeepMind calls "subconscious plagiarism" [8]. Lean checks logic, not meaning or novelty.

3 Scoring methodology

3.1 Impact Score

Each problem is rated 1–10 on five criteria, and the weighted sum is scaled to 0–100:

$$I = 10\,(0.30\,C + 0.25\,R + 0.15\,L + 0.15\,N + 0.15\,X)$$

Table 3: Impact Score criteria.

Symbol	Criterion	High score means
C	Centrality	The statement sits at the core of a major field
R	Downstream reach	Many theorems are conditional on it or would follow from it
L	Longevity and attention	Open for decades despite serious, well-known attempts
N	Expected method yield	A solution would likely require ideas reusable elsewhere
X	External impact	Consequences for physics, computer science, cryptography or engineering

Centrality and reach dominate because a theorem's value to mathematics is mostly what it unlocks. Collatz scores low ($I = 41$) despite fame, since little depends on it; P vs NP scores highest ($I = 98.5$).

3.2 Compute tiers

We express cost as the expected number of **output tokens** an AI-only program would need to resolve the problem, using published runs as anchors.

Table 4: Compute tiers and the runs that anchor them.

Tier	Output tokens	Typical form	Observed anchors
A: Sweep and search	10^6–10^7	Single prompts, evolutionary search with a verifier	Schubitope counterexample, first prompt, 38 min [21]; AlphaProof Nexus, a few hundred dollars per Erdős problem [4]; OpenAI's Oct 2026 results averaged ~3 hours of ChatGPT Pro thinking each [12]
B: Agent run	10^7–10^8	One model, hours to a day, subagents and Lean	Crouzeix, 16-hour run [18]; zeta bound, 31M tokens [11]

Tier	Output tokens	Typical form	Observed anchors
C: Small swarm	10^8–10^{10}	Tens to hundreds of agents over days	Euler blow-up, ~100 agents for ~50 hours [5]
D: Mega-swarm	10^{10}–10^{11}	Thousands of agents, coordinated	Navier–Stokes (forced), ~130B tokens, 88 hours [5]
E: Beyond scale	$> 10^{11}$, uncertain	Requires theory that may not exist yet	No precedent

3.3 Priority Index

Let t be the estimated $\log_{10}$ of output tokens and p the estimated probability that an AI-only program resolves the problem (in its standard form) within 12 months at that budget. We define

$$\Pi = \frac{I \cdot p}{t - 5},$$

the expected impact per order of magnitude of compute above a 10^5-token baseline. Π rewards cheap, likely wins, and it is deliberately harsh on moonshots: a program that only maximizes Π never attempts the Riemann Hypothesis. Section 10 shows how to budget for both.

3.4 Limitations

p and t are forecasts with wide error bars, and the 2026 record shows they drift quickly downward: problems this paper places in Tier C may be Tier B by early 2027. Impact weights are a choice; a number theorist and a combinatorialist would set them differently. The ranking is most useful as an *ordering*, not as precise values.

4 Tier A: The cheapest wins (10^6–10^7 tokens)

These problems share one property: a candidate answer can be verified by a program in seconds. They suit paradigm 1 (evolutionary search) and paradigm 3 at low budget.

Table 5: Tier A problems.

Problem	Open since	Verifier	I	t	p	Π
Remaining Epoch Open Problems (constructions)	various	Bespoke Epoch verifiers	30.0	6.5	0.50	10.0
Kissing numbers and packings, dimensions 9–31	1960s+	Exact geometry	40.5	7.0	0.50	10.1
Hadamard matrices: next open orders	1893 (Hadamard)	Matrix product check	34.5	7.0	0.50	8.6
Small Ramsey numbers, e.g. $R(5,5)$	1955	Graph check (lower); SAT/Lean (upper)	42.0	8.5	0.20	2.4

Problem	Open since	Verifier	I	t	p	Π
Hadwiger–Nelson: chromatic number of the plane	1950	Unit-distance graph colouring	39.0	8.0	0.15	2.0

Remaining Epoch Open Problems. Thirty-nine of the 49 listed problems are unsolved, most of them finite constructions or record-breaking objects (Diophantine solutions, curves with many rational points, knot invariants) [16]. They are designed to be machine-checked, so the only cost is search. *Plan:* run an evolutionary loop per problem with the verifier as fitness, then escalate to an agent for a construction family.

Kissing numbers and packings. AlphaEvolve raised the 11-dimensional kissing number from 592 to 593 [7], and a later open-world agent system reported exact 604-point configurations in dimension 11 [22]. Every improved configuration is a publishable record and can be checked exactly. Impact is moderate (coding theory, communications), and p is high for "improve at least one dimension".

Hadamard matrices. The order-668 matrix was found in 2026 by a team crediting Claude [16]. The next unknown orders are pure search targets with an instant verifier.

Small Ramsey numbers. $R(5,5)$ is known to lie in $[43, 46]$. Raising the lower bound needs one good colouring, a pure construction; lowering the upper bound needs an exhaustive argument, which is a SAT-plus-Lean engineering problem. We place it at the top of Tier A because closing it outright likely needs both.

Hadwiger–Nelson. The plane needs between 5 and 7 colours; the lower bound of 5 came from a computer-found graph in 2018. Raising it to 6 is a construction problem (a finite unit-distance graph that is not 5-colourable), which is exactly the shape AI search excels at. Lowering 7 requires a new colouring of the whole plane, which is harder.

Why start here. Tier A has the highest Π of any tier, produces verifiable outputs with no fidelity risk, and builds the infrastructure (verifiers, search loops, Lean pipeline) that every later tier needs. Its weakness is impact: each result is a record, rarely a theory.

5 Tier B: Named conjectures with mature toolkits (10^7–10^8 tokens)

These are named problems where the literature already contains most of the machinery, and a solution likely recombines known techniques in a new way: the profile of Sendov, Crouzeix and Chvátal in 2026.

Table 6: Tier B problems. *For FME, p is the probability of resolving at least five more of the 68.

Problem	Open since	Status of best partial result	I	t	p	Π
Bloom's 68 hard open Erdős problems (FME)	1930s–1990s	2 of 68 resolved by AI with Lean [20]	51.5	8.0	0.60*	10.3
Union-closed sets (Frankl)	1979	Constant raised to 0.38288 in Oct 2026; conjecture is 1/2 [21]	49.0	8.0	0.20	3.3

Problem	Open since	Status of best partial result	I	t	p	Π
Sunflower conjecture (Erdős–Rado)	1960	$(c \log k)^k$-type bounds since 2019	67.0	9.0	0.10	1.7
Lonely runner (all n)	1967	Proved up to 14 runners, computer-assisted [23]	49.0	8.5	0.10	1.4
Graceful tree (Ringel–Kotzig)	1963	Known for many tree classes	40.5	8.5	0.10	1.2

The 68 hard Erdős problems. Thomas Bloom chose these as both significant and difficult; the rest of the 652 open Erdős problems include many that are merely neglected [20]. They come pre-formalized in Lean, which removes fidelity risk, and two were cracked by a pre-release model in September. A program that runs a Tier B agent on each of the 68 is the single best use of a 10^8–10^9 token budget.

Union-closed sets. Gilmer's 2022 entropy method broke the problem open, and successive improvements have pushed the constant toward Liu's natural barrier near 0.382; an October 2026 preprint used ChatGPT to pass it [21]. Reaching 1/2 likely needs a genuinely new idea, which lowers p, but the incremental ladder (each new constant is publishable) keeps compute productive.

Sunflower. The highest-impact problem in Tier B, because sunflower bounds feed circuit lower bounds and cryptography. The 2019 Alweiss–Lovett–Wu–Zhang breakthrough and its refinements give a modern toolkit. Its expected cost sits at the top of the tier.

Lonely runner. The case-by-case frontier is now 14 runners via certified finite checks [23]. The general case is a Diophantine-approximation statement where a structural idea, not more cases, is needed.

Graceful tree. Famous, easy to state, and checkable for any given tree, but with low downstream reach. A good test of whether AI can find a uniform labelling rule rather than case analysis.

6 Tier C: Major conjectures needing new structure (10^8–10^{10} tokens)

At this tier a solution probably requires an idea no paper contains yet. The 2026 precedents are the Jacobian disproof, the linear Hadwiger proof (found by GPT-6 Astra following directions from Norin and Steiner [24]), and Kakeya in $\mathbb{R}^4$, if OpenAI's claim survives review [12].

Table 7: Tier C problems.

Problem	Open since	What is known	I	t	p	Π
Kakeya conjecture in $\mathbb{R}^n$, $n \geq 5$	1917 (Kakeya); 1971	$n = 3$ proved (Wang–Zahl 2025); $n = 4$ claimed Oct 2026 [12]	75.5	10.0	0.10	1.5
Jacobian conjecture, dimension 2	1939	False for $n \geq 3$ (2026) [10]	64.5	9.5	0.08	1.1
Erdős–Hajnal conjecture	1989	Many forbidden-graph cases	64.5	9.5	0.07	1.0

Problem	Open since	What is known	I	t	p	Π
Hadwiger's conjecture (full)	1943	Linear version proved Oct 2026 [24]	64.5	9.5	0.05	0.7
Littlewood conjecture	c. 1930	Exceptional set has dimension 0 (Einsiedler–Katok–Lindenstrauss)	56.0	10.0	0.05	0.6
Hadamard conjecture (all $n = 4k$)	1893/1933	Known for many families; 668 found 2026	55.0	10.0	0.05	0.6

Kakeya in higher dimensions. The highest-impact problem in Tier C: Kakeya sits beneath restriction theory and Bochner–Riesz in harmonic analysis. With $n = 3$ proved and $n = 4$ claimed, the natural question is whether the four-dimensional argument generalizes. If OpenAI's proof holds, p rises sharply; it is the clearest case where reading the newest AI proofs is itself the attack plan.

Jacobian conjecture in dimension 2. The 2026 counterexamples live in dimension 3 and up [10]. The plane case is now the whole problem. If it is false, a counterexample is a Tier A-style search over polynomial maps, which is why p is not negligible; if it is true, it needs real algebraic geometry. A dual attack (search for counterexamples and seek structure in parallel) is the right design.

Erdős–Hajnal and Hadwiger. Both are central in structural graph theory and both saw 2026 progress on weaker forms. A disproof of Hadwiger in high t would be a construction, matching R2. A proof likely needs a new decomposition theorem.

Littlewood and the Hadamard conjecture. Both demand a uniform statement over infinitely many cases where computation only supplies evidence. They are lower in Π because no verifier-friendly sub-step is known to carry the full result.

7 Tier D: Famous problems with deep obstructions (10^{10}–10^{11} tokens)

These problems have known *barriers*: theorems explaining why existing methods cannot work. A solution must route around them.

Table 8: Tier D problems.

Problem	Open since	Known barrier or best result	I	t	p	Π
Navier–Stokes, unforced blow-up or regularity	1934 / 2000	Forced Clay case claimed Sept 2026; method provably needs the force [6]	82.0	11.0	0.05	0.7
abc conjecture (accepted proof)	1985	Mochizuki's claimed proof not accepted by most experts	81.0	11.0	0.02	0.3
Twin prime conjecture	antiquity / 1849	Bounded gaps (246); parity barrier	73.0	11.0	0.02	0.2
Binary Goldbach	1742	Ternary proved (Helfgott 2013); parity barrier	55.0	11.0	0.02	0.2

Problem	Open since	Known barrier or best result	I	t	p	Π
Collatz	1937	"Almost all" orbits (Tao 2019)	41.0	11.0	0.01	0.1

Unforced Navier–Stokes. The most promising Tier D target, for a reason specific to AI. OpenAI's swarm first resolved the *unforced* Euler blow-up, using about 100 agents for 50 hours [5], and blow-up is a construction problem in the sense of R2. If unforced Navier–Stokes blows up, a construction may exist and swarms are good at finding them. If it does not, a regularity proof is theory, and theory is where AI is weakest [6]. The external impact (fluid physics, turbulence) is the highest of any problem outside Tier E.

abc. A uniquely AI-shaped opportunity: the community dispute is about whether a long proof is correct. A full Lean formalization of inter-universal Teichmüller theory, or an independent proof, would settle it either way. Formalizing Perelman's Poincaré proof in Lean was reported in September 2026 [21], which shows that large-scale formalization of disputed or complex proofs is now feasible.

Twin primes and Goldbach. Both hit the *parity barrier* in sieve theory: sieves alone cannot distinguish numbers with an odd number of prime factors from those with an even number. Progress needs a new source of arithmetic information. Good partial targets exist (shrinking the 246 gap, exceptional-set bounds for Goldbach).

Collatz. Included as a calibration point. It has the highest fame-to-impact ratio on this list. Erdős reportedly said mathematics is "not ready" for such problems, and little else depends on the answer. It is a poor use of compute.

8 Tier E: The hardest problems ($> 10^{11}$ tokens, uncertain)

Table 9: Tier E problems.

Problem	Posed	Best partial result	I	t	p	Π
P vs NP	1971	Relativization, natural-proofs and algebrization barriers	98.5	12	0.005	0.07
Riemann Hypothesis	1859	At least 67.2% of zeros on the line (Claude, 2026) [11]	94.0	12	0.010	0.13
Yang–Mills existence and mass gap	2000 (Clay)	Constructive QFT in lower dimensions	89.0	12	0.005	0.06
Birch and Swinnerton-Dyer	1960s	Known for analytic rank ≤ 1	85.5	12	0.010	0.12
Hodge conjecture	1950	Known for divisors (Lefschetz (1,1)) and some cases	77.0	12	0.010	0.11

Why these sit at the top of impact and the bottom of priority. Each requires a theory

that does not yet exist, has no verifier that can check a candidate solution's core idea in seconds, and resists the construction strategies that dominate AI's record. Anthropic's own assessment of its zeta-zero result is instructive: it is a real improvement, but the company does "not expect that the techniques Claude used will lead to proving the Riemann hypothesis" [11]. The one Tier E problem with a construction-shaped answer is P vs NP *if* P = NP, which most experts consider unlikely.

How to spend compute on Tier E anyway: partial-progress ladders. Each problem has a sequence of publishable intermediate targets that are verifier-friendly or at least well-specified:

- **Riemann:** raise the critical-line proportion beyond 67.2%; widen the zero-free region; improve zero-density estimates beyond Guth–Maynard (2024); tighten the de Bruijn–Newman constant bounds.
- **P vs NP:** explicit circuit lower bounds for restricted classes; the 2026 Claude result refuting the 3SUM and APSP hypotheses [21] shows that fine-grained complexity conjectures can fall to algorithm search, so they are good probes of which hardness assumptions are sound.
- **BSD:** extend the rank ≤ 1 results; large-scale numerical verification for families of curves; new cases of the p-adic or Iwasawa main conjectures.
- **Hodge:** new families of varieties where it holds; the Mumford–Tate conjecture for hyper-Kähler varieties was reported proved with AI help in October 2026 [21], a sign that adjacent conjectures are moving.
- **Yang–Mills:** rigorous constructions in lower dimensions or on lattices with controlled continuum limits.

9 Master ranking

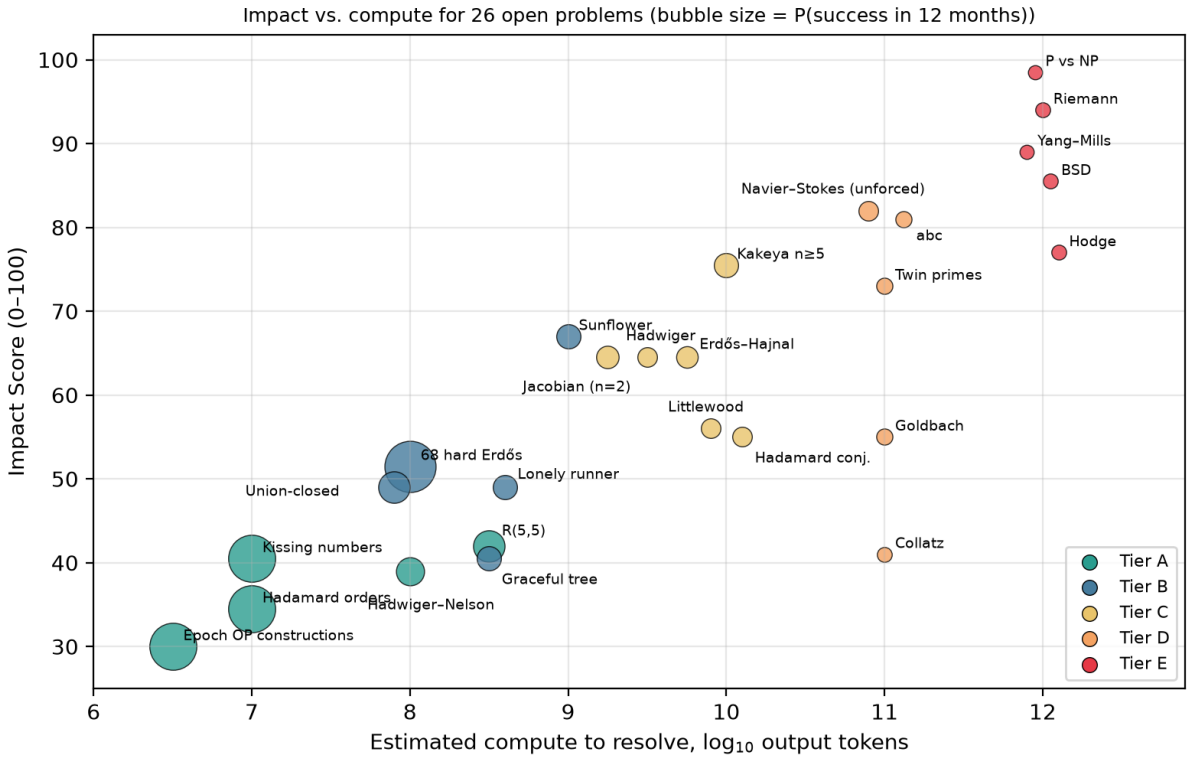


Figure 1: Impact vs. estimated compute for 26 open problems. Bubble size is the estimated probability of success within 12 months.

Table 10: All 26 problems ranked by Priority Index.

Rank	Problem	Tier	I	$\log_{10}$ tokens	p	Π
1	Bloom's 68 hard Erdős problems	B	51.5	8.0	0.60	10.30
2	Kissing numbers and packings	A	40.5	7.0	0.50	10.12
3	Remaining Epoch Open Problems	A	30.0	6.5	0.50	10.00
4	Hadamard matrices: next orders	A	34.5	7.0	0.50	8.62
5	Union-closed sets	B	49.0	8.0	0.20	3.27
6	$R(5,5)$ and small Ramsey numbers	A	42.0	8.5	0.20	2.40
7	Hadwiger–Nelson	A	39.0	8.0	0.15	1.95
8	Sunflower conjecture	B	67.0	9.0	0.10	1.68
9	Kakeya, $n \geq 5$	C	75.5	10.0	0.10	1.51
10	Lonely runner	B	49.0	8.5	0.10	1.40
11	Graceful tree	B	40.5	8.5	0.10	1.16
12	Jacobian, dimension 2	C	64.5	9.5	0.08	1.15
13	Erdős–Hajnal	C	64.5	9.5	0.07	1.00

Rank	Problem	Tier	I	$\log_{10}$ tokens	p	Π
14	Hadwiger (full)	C	64.5	9.5	0.05	0.72
15	Navier–Stokes, unforced	D	82.0	11.0	0.05	0.68
16	Littlewood	C	56.0	10.0	0.05	0.56
17	Hadamard conjecture	C	55.0	10.0	0.05	0.55
18	abc (accepted proof)	D	81.0	11.0	0.02	0.27
19	Twin primes	D	73.0	11.0	0.02	0.24
20	Binary Goldbach	D	55.0	11.0	0.02	0.18
21	Riemann Hypothesis	E	94.0	12.0	0.010	0.13
22	Birch and Swinnerton-Dyer	E	85.5	12.0	0.010	0.12
23	Hodge	E	77.0	12.0	0.010	0.11
24	Collatz	D	41.0	11.0	0.010	0.07
25	P vs NP	E	98.5	12.0	0.005	0.07
26	Yang–Mills	E	89.0	12.0	0.005	0.06

Reading the ranking. Three findings stand out. First, the top four are all verifier-backed, consistent with R1. Second, Kakeya ($n \geq 5$) is the best high-impact bet, because a fresh lower-dimensional proof gives the program a template. Third, unforced Navier–Stokes is the only Tier D problem in the top 15, because its likely answer (blow-up) has a construction shape.

10 Proposed research program

10.1 Principles

1. **Climb the ladder.** Spend early compute where Π is highest and use the results to build tooling.
2. **Always attack both directions.** For every conjecture, run a counterexample search alongside a proof attempt; 2026's biggest results were disproofs.
3. **Reserve a moonshot budget.** Pure Π-maximization never attempts Tier E. Fix 10–15% of compute for partial-progress ladders on Tier D and E.
4. **Verify, then verify the statement.** Every claim gets a Lean proof where possible, and a human-readable statement audit regardless.

10.2 Phases and budgets

Table 11: A phased program. Budgets are cumulative orders of magnitude per phase.

Phase	Duration	Targets	Budget (output tokens)	Expected outputs
0: Infrastructure	2 weeks	Verifiers, Lean pipeline, literature-search agent	10^7	Reusable harness
1: Sweep	1 month	Tier A (all 5), obscure Erdős and OEIS problems	10^8	10–30 records and small resolutions

Phase	Duration	Targets	Budget (output tokens)	Expected outputs
2: Agents	2 months	Tier B (all), FME 68	10^9	3–8 named results
3: Swarms	3 months	Kakeya $n \geq 5$, Jacobian $n = 2$, Erdős–Hajnal, Hadwiger	10^{10}	0–2 major results
4: Moonshots	ongoing, 10–15% of compute	Unforced Navier–Stokes; Riemann and BSD ladders; abc formalization	10^{10}–10^{11}	Partial-progress papers

10.3 Harness design

- **Generator layer:** a frontier reasoning model, plus a cheap open model for high-volume evolutionary search (open 8B models have improved AlphaEvolve-era bounds).
- **Critic layer:** independent models referee each other's arguments; the zeta-zero run used 13 of its 60 subagents as validators [11].
- **Verifier layer:** problem-specific checkers for constructions; Lean for proofs; a statement-fidelity check comparing the formal statement with the literature version.
- **Novelty layer:** an agent that searches the literature before any claim is reported, to catch rediscoveries (R3, R5); Anthropic's run downloaded 54 arXiv papers for this purpose [11].
- **Coordinator:** cross-pollinates promising partial ideas between agent groups, the step OpenAI credits for its Navier–Stokes result [5].

10.4 Disclosure protocol

Follow the September 2026 recommendations of the Advisory Group on Mathematics and AI at the Institute for Advanced Study: publish the model, the exact prompt and the compute behind each result [12]. Record the human role honestly, using Epoch's "AI", "human + AI" and "human" labels [16]. Never announce a result before the statement audit and an outside expert's reading.

11 Risks and open questions

- **Statement-fidelity failures.** The Navier–Stokes episode shows a correct proof of the wrong formulation can still win the headline [6]. Mitigation: audit every formal statement against the problem's accepted intent, not only its letter.
- **Verification capacity.** Mathematicians cannot referee hundreds of AI proofs a month; Tao has publicly criticized the "insane" pace [12]. Lean helps with logic but not with understanding. A program that floods the field may reduce trust in its own results.
- **Shrinking novelty window.** If OpenAI's 372 results include several problems in this paper, the ranking must be recomputed. Re-checking status is a standing task, not a one-off.
- **Value of understanding.** A solved problem that no human understands has lower N (method yield) than its score assumes. Investing in exposition (Tao-style "digestions") is part of the work.
- **Does compute keep buying depth?** R4 is an empirical trend over five orders of magnitude, not a law. Tier E may need ideas that no amount of search produces.

12 Conclusion

AI-driven pure mathematics in October 2026 is lopsided. It is superhuman at finding explicit objects that a computer can check, and very strong at combining known techniques under long autonomous runs. It is still weak at inventing the new theory that the deepest problems require. The right research program follows that shape: sweep the verifier-backed problems first (Tier A), run long agents on mature named conjectures (Tier B), spend swarm compute where a fresh proof gives a template (Kakeya) or where the answer is likely a construction (unforced Navier–Stokes), and treat the Riemann Hypothesis, P vs NP and the other Millennium Problems as ladders of partial results rather than single targets. By this paper's scoring, the best mathematics per token today is in Bloom's 68 hard Erdős problems, and the best high-impact bet is Kakeya in five or more dimensions.

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